A Hybrid Quantum-Classical Attention Mechanism for Efficient Large Language Models

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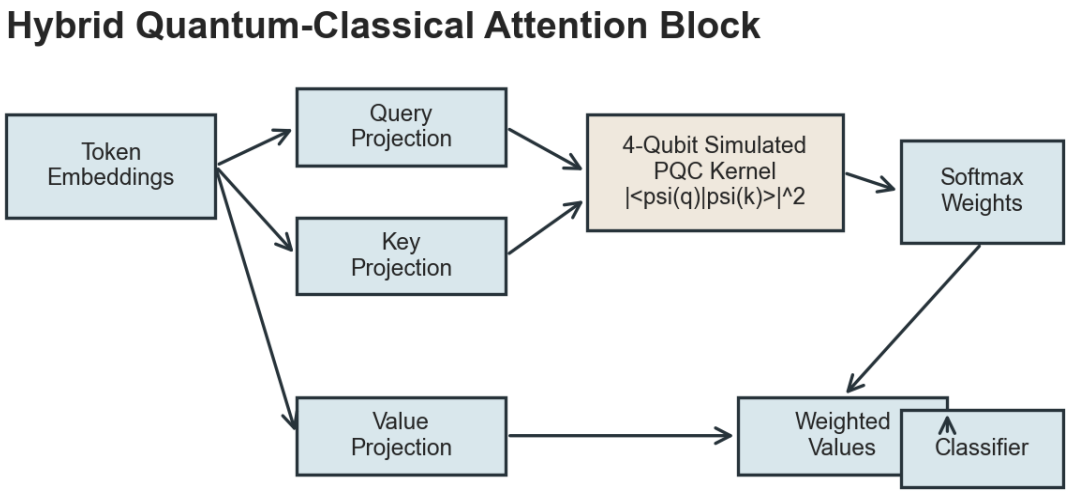
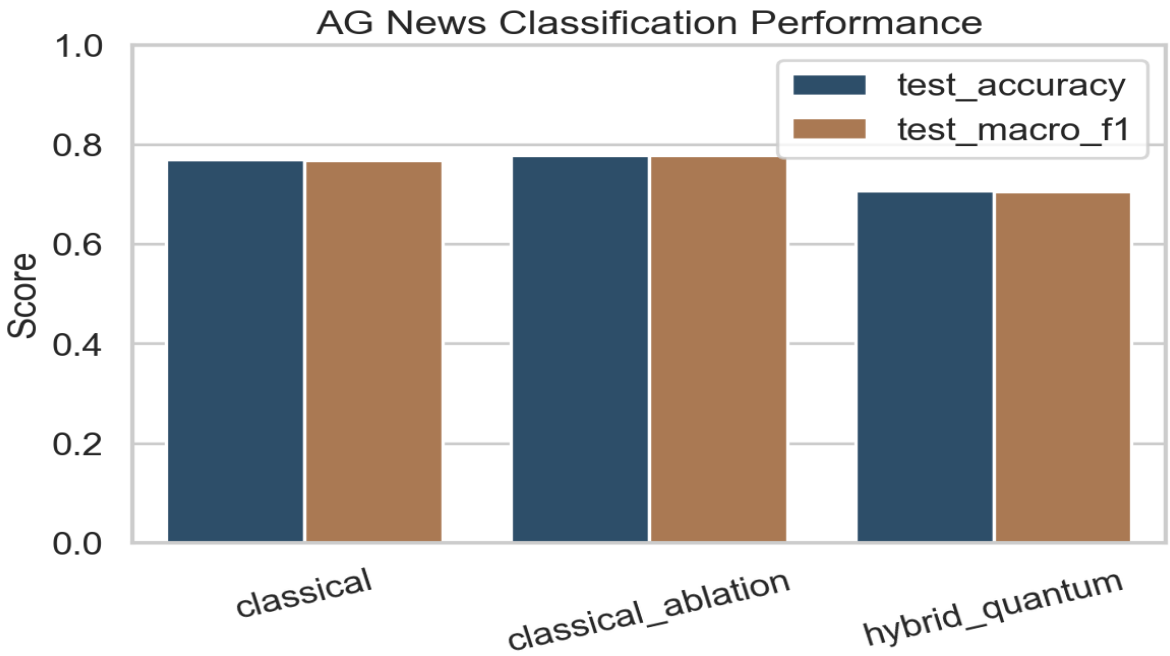
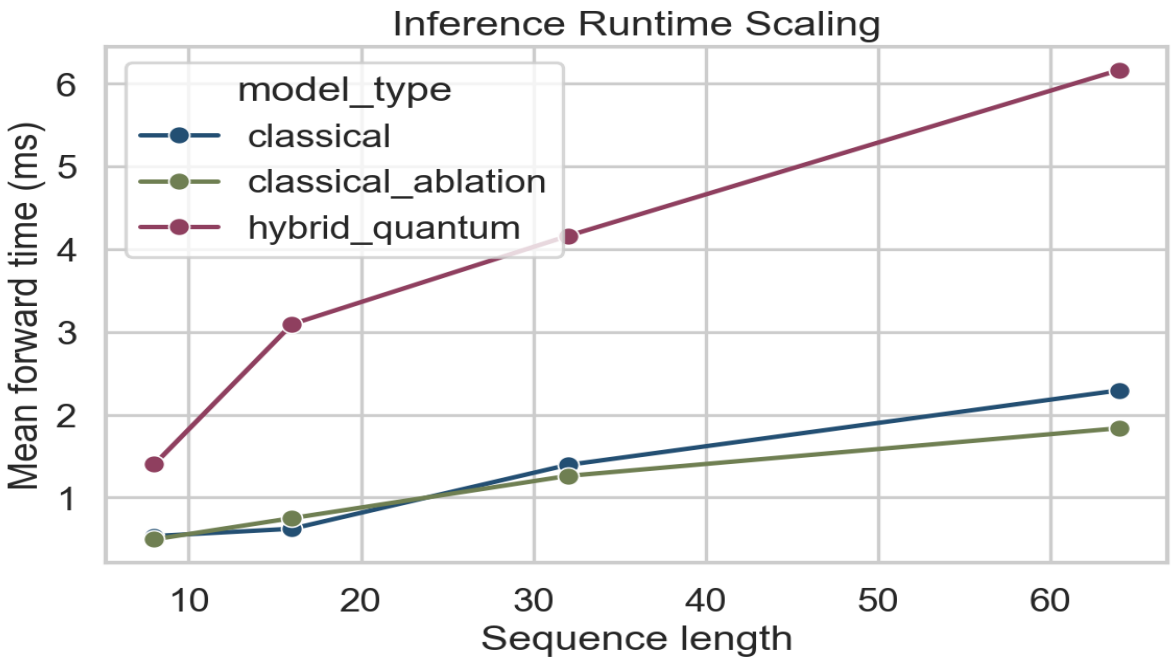
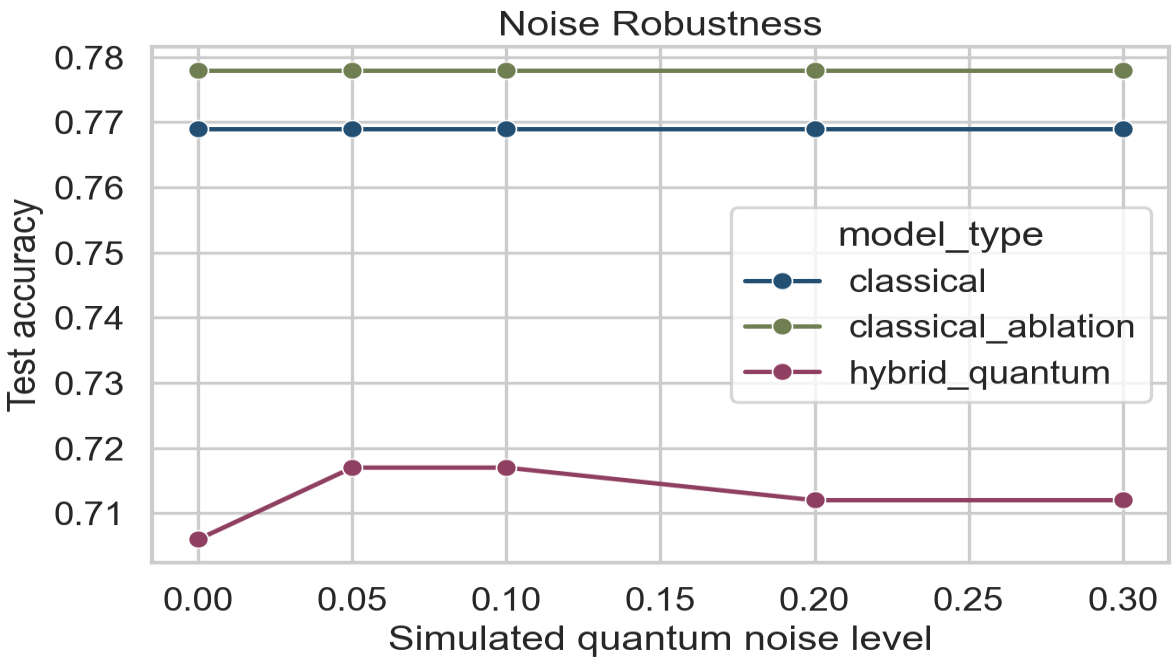
Abstract

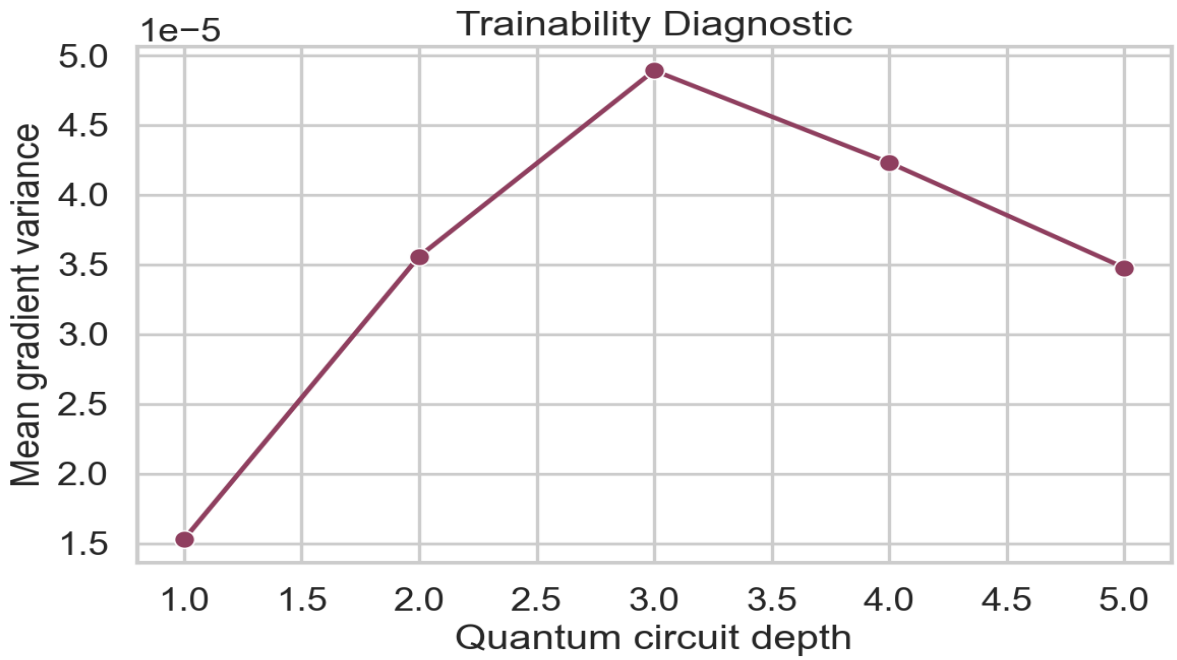
This compact study replaces only the query-key dot product in attention with a simulated four-qubit quantum kernel. On AG News, the classical, hybrid, and ablation models reached 76.9%, 70.6%, and 77.8% test accuracy. The hybrid model was 2.99x the classical runtime at sequence length 32, showing no classical-simulation speedup but providing a reproducible diagnostic framework.

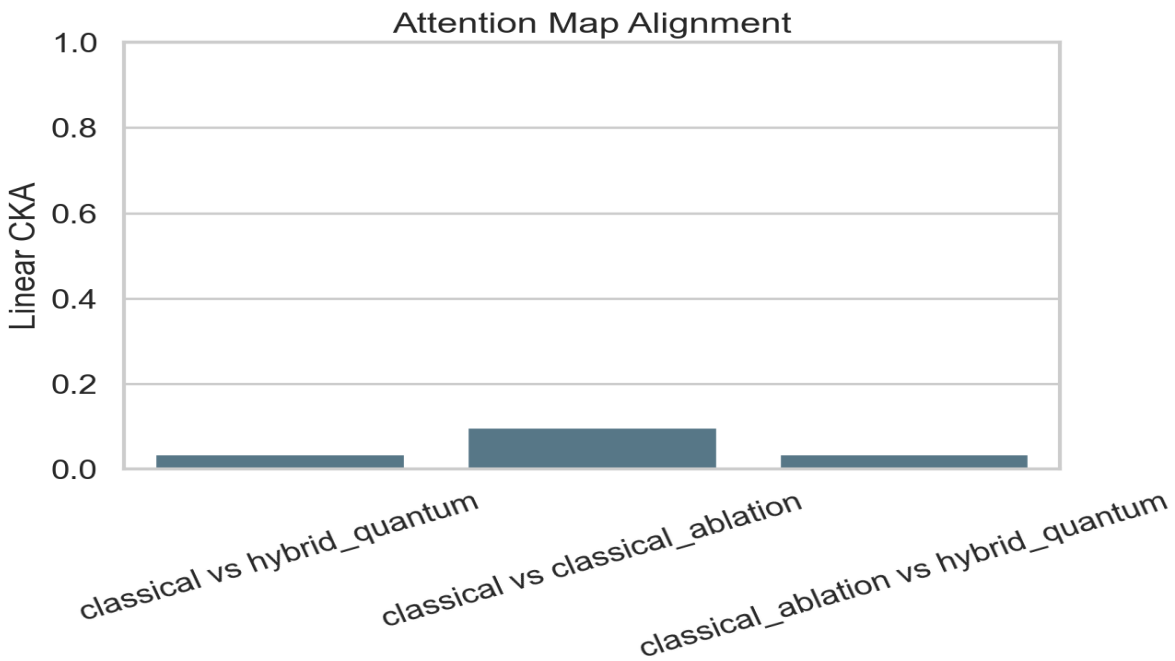





Inference Runtime Scaling
model_type
classical
classical_ablation
hybrid_quantum
Mean forward time (ms)
1
2
3
4
5
6
10
20
30
40
50
60
Sequence length


Noise Robustness
model_type
classical
classical_ablation
hybrid_quantum
Test accuracy
0.71
0.72
0.73
0.74
0.75
0.76
0.77
0.78
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Simulated quantum noise level





Interpretation

The result should be read as a feasibility and diagnostic study. The hybrid circuit can be integrated into attention and compared against matched baselines, but state-vector simulation adds overhead. The strongest contribution is the controlled experiment design: accuracy, runtime, CKA, gradient variance, and noise robustness are all measured from saved metric files.

References

Vaswani et al. (2017); Li et al. (2023), arXiv:2205.05625; Chen et al. (2025), Neural Networks 185; Kornblith et al. (2019), ICML; McClean et al. (2018), Nature Communications; Shen et al. (2025), arXiv:2402.10787.